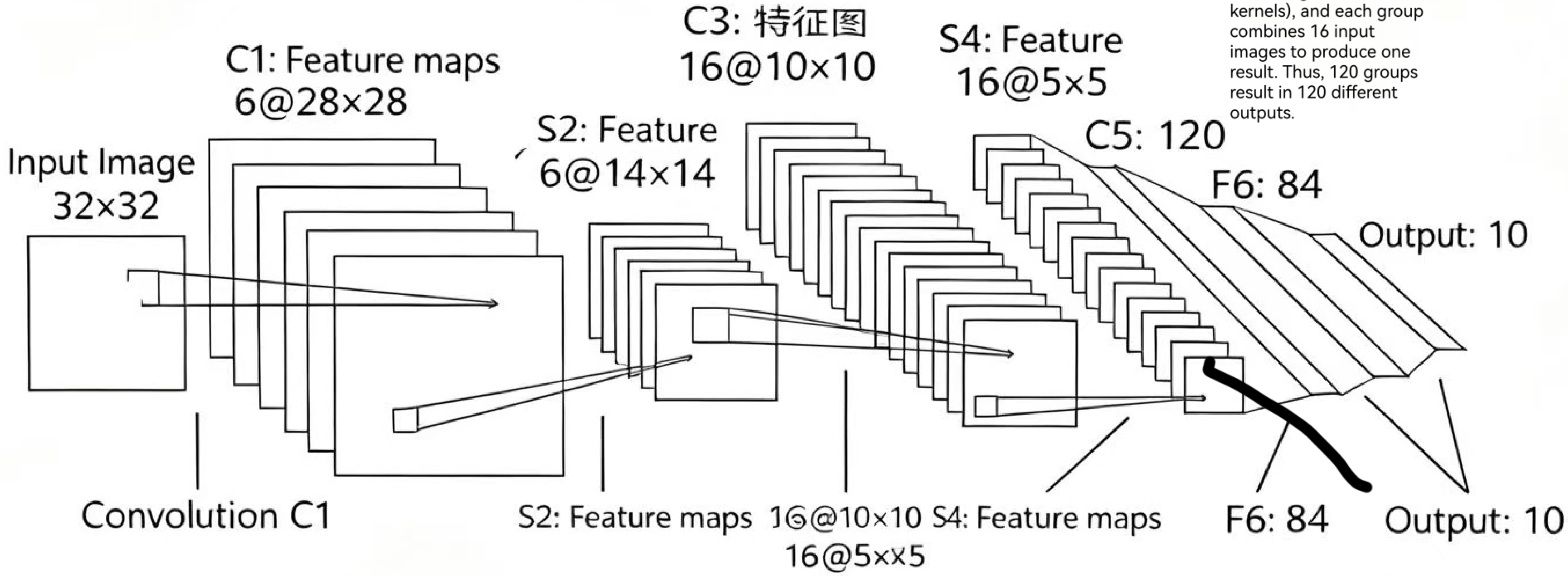
C5 There are 120 different convolution kernels (each group consisting of 16 5×5 kernels), and each group combines 16 input images to produce one result. Thus, 120 groups result in 120 different outputs.



input layer
size: 32 * 32 image

C1 layer (convolutional layer)
6 5 * 5 convolutional kernals are selected, with stride = 1. This result in 6 feature maps of size 28 * 28 (32 - 5 -1), and the numer of neurons is 6 * 28 * 28 = 4704

Layer S2 (Downsampling Layer): The 6 28×28 feature maps obtained from C1 undergo mean pooling. The pooling kernel size is set to 2×2, and the stride is 2. The obtained means are multiplied by a weight and a bias, and then used as the input for the Sigmoid activation function, resulting in 6 14×14 feature maps. The number of neurons is 6×14×14 = 1176.

C3 layer (convolutional layer): 16 groups of 5×5 convolution kernels (the number of convolution kernels in the first 6 groups is 3, the middle 6 groups are 4, the following 3 groups are 4, and the last 1 group is 6) are used to perform convolution on the feature maps output by the S2 layer. After adding bias and applying the activation function (Sigmoid), 16 new feature maps of size 10×10 (14-5+1=10) are obtained. At this time, the number of neurons is 16×10×10 = 1600.

S4 layer (downsampling layer) perform max pooling on 16 10*10 eature maps with 2*2 pooling kernal and the stride = 2 + bias as the input of sigmod funtion. we can get 16 5*5 feture maps. the number of neurons is 16*5*5 = 400

The F6 layer uses 84 neurons to fully connect these 120 values, resulting in 84 outputs, which are used to represent 7×12 character images.

$$d_i = \sum_{j=1}^{84} [x_j - w_{ij}]^2$$

$$\arg\min_{i \in \{0,1,\dots,9\}} d_i$$